Date	03 July 2026
Team ID	SWTID–2026–4593
Team Size	3
Team Leader	Hari Bellamkonda
Team Member 1	Kesha Vardhan Chinni
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Project Name	Machine Learning–Based Credit Card Approval Prediction System
Maximum Marks	4 Marks

Problem–Solution Fit

Introduction

The Problem–Solution Fit explains how the proposed Machine Learning–Based Credit Card Approval Prediction System directly addresses the challenges faced by financial institutions during the credit card approval process. It demonstrates how the proposed solution satisfies stakeholder needs by providing faster, more accurate, secure, and intelligent approval predictions.

Problem	Impact	Proposed Solution	Expected Benefit
Manual credit card application screening is time-consuming.	Application processing is delayed, increasing operational workload.	Automate application screening using Machine Learning classification models.	Faster approval decisions with reduced manual effort.
Human decision-making may introduce inconsistency and bias.	Applicants with similar financial profiles may receive different outcomes.	Use trained Machine Learning models to ensure consistent prediction based on historical data.	Improved fairness, consistency, and decision accuracy.
Large volumes of applications are difficult to process efficiently.	Banks experience reduced productivity and increased processing costs.	Develop a scalable web-based prediction platform integrated with an optimized Machine Learning model.	High-speed processing with improved operational efficiency.
Applicants do not know their eligibility before submitting applications.	High rejection rates and poor customer experience.	Provide an online eligibility prediction system that estimates approval likelihood before formal application.	Better customer awareness and fewer unnecessary application submissions.
Sensitive applicant information requires secure handling.	Risk of data breaches and privacy violations.	Implement authentication, role-based access control, encrypted communication, and secure database storage.	Enhanced security, confidentiality, and regulatory compliance.

Problem–Solution Fit Summary

- Automates credit card approval prediction.
- Improves prediction accuracy using Machine Learning.
- Reduces application processing time.
- Enhances customer experience.
- Supports secure handling of applicant information.
- Enables scalable deployment through Flask and cloud technologies.
- Assists financial institutions in making reliable credit decisions.

Conclusion

The proposed Machine Learning–Based Credit Card Approval Prediction System effectively addresses the limitations of traditional manual credit evaluation by combining intelligent prediction algorithms, secure web technologies, and automated workflows. The solution provides faster, more consistent, and reliable credit approval decisions while improving operational efficiency and customer satisfaction.